

Dengue Outbreak Risk Prediction

Predicting subzone-level dengue cluster risk —
strategic batch + tactical real-time inference, served from one model

330+

SG Subzones scored

7 yrs

Real cluster data

100%

Gov open data

Dengue control is a two-speed problem

Singapore's 2020 outbreak — 35,000+ cases — exposed two distinct operational gaps. Conflating them with one model is the strategic mistake.

STRATEGIC LOOP · WEEKLY BATCH

Where should fogging be deployed next week?

Cadence	Weekly · Monday 06:00 SGT
User	NEA Ops + Pest Control Operators
Action	Plan fogging crews + equipment staging
If wrong	Missed fogging run — recoverable

Latency tolerance: days

TACTICAL LOOP · EVENT-DRIVEN REAL-TIME

Who needs same-day outreach right now?

Cadence	Event-driven · per case notification
User	CDC Officer at MOH
Action	Triage: home visit, advisory, testing van
If wrong	Vulnerable resident → WHO days 4-7 critical phase

Latency tolerance: sub-second

Two cost functions · two cadences · one underlying model serving both

The dataset

ANCHOR

SGCharts Dengue Cluster Snapshots

May 2013 → Nov 2020 · 250+ bi-weekly snapshots · Community CSV

- Lat/long cluster boundaries + case counts
- 7 years of real spatial cluster data
- Covers 2020 record outbreak (35k+ cases) — concept drift event

AUXILIARY · WEATHER

MSS Weather Station Records

Met Service Singapore · station-level daily · 2013–present

- Rainfall, temperature, relative humidity per station
- Replaces NEA realtime API — extends usable training window
- Drives lag features: 1w / 2w / 4w rainfall, 4w temp mean

AUXILIARY · DEMOGRAPHICS + GEOMETRY

SingStat Census + URA Master Plan

Census 2020 · URA Master Plan 2019 GeoJSON

- Subzone-level population, age, comorbidity proxies
- Vulnerability index (% elderly + chronic) for tactical layer
- URA polygons → spatial join target for label derivation

LABEL

Spatial Join → Binary Label

Derived from anchor · fully transparent process

- URA subzones intersected with SGCharts cluster polygons
- Label = 1 if active cluster in subzone in next 14 days
- No synthetic data — every label traces to real cluster

Users and how they consume the system

NEA Operations Team

STRATEGIC LOOP · PRIMARY USER

When	Weekly — every Monday morning
Receives	Risk map: 330 subzones colour-coded Low/Med/High
Action	Pre-deploy fogging teams to High-risk subzones

Pest Control Operators

STRATEGIC LOOP · SECONDARY USER

When	Weekly — before scheduling fogging rounds
Receives	Ranked list of high-risk subzones with confidence
Action	Prioritise contracts, allocate crews to predicted hotspots

CDC Officer at MOH

TACTICAL LOOP · NEW USER

When	Event-driven — per confirmed case notification
Receives	Real-time alert: score, vulnerability features, 200m radius
Action	Triage: home visit, advisory, testing van dispatch

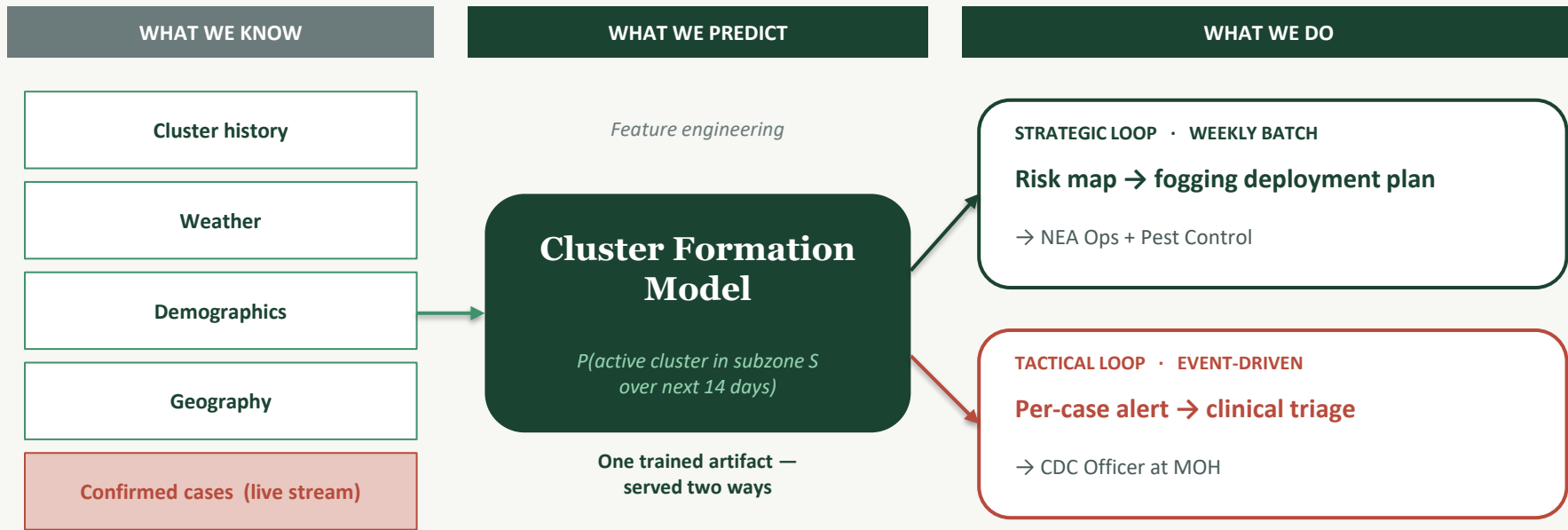
Internal Data / Ops Team

INTERNAL · TOOLING ONLY

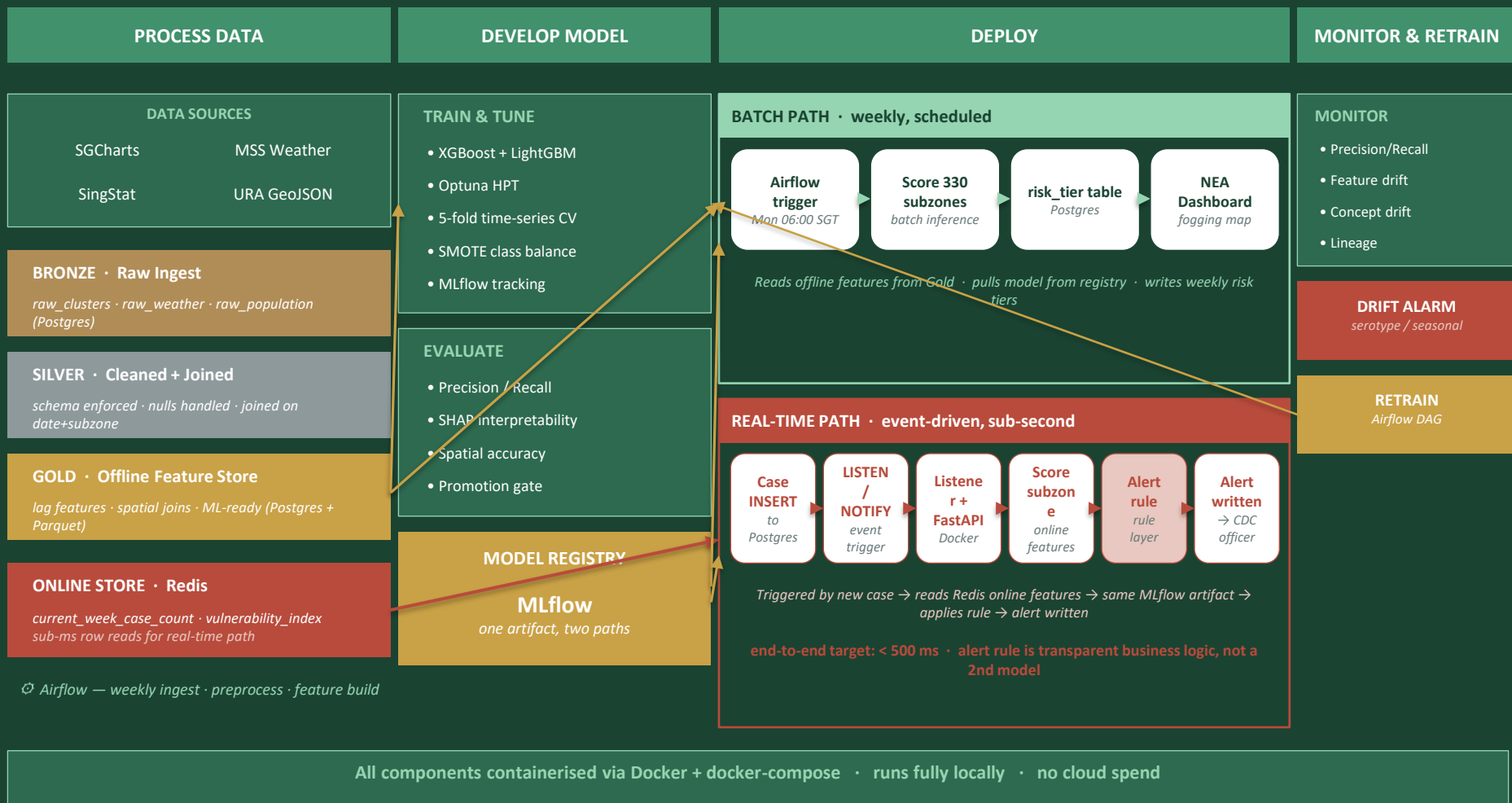
When	Ongoing — weekly monitoring + drift alerts
Receives	Model performance dashboard: precision, recall, drift
Action	Review pipeline health; trigger retraining when Gini drops

One model, two inference loops

Shared predictive logic; different feature freshness, different decision thresholds, different downstream actions.



Same predictive logic. Different freshness contracts. Different thresholds. Different actions.



Design choices & what we decided

DESIGN CHOICES · RATIONALE

Two-layer by design — batch + real-time

Different operational loops, different cost functions. Same trained model, two inference paths.

XGBoost / LightGBM, not deep learning

Tabular spatial+temporal data. Interpretable via SHAP. Trains in minutes locally.

Time-series sliding-window CV

Standard k-fold leaks future cluster data — temporal CV is the only valid choice.

Recall ≥ 0.70 on High-risk tier

Asymmetric cost: false negative = missed outbreak. We optimise for the more expensive error.

WHAT WE DECIDED SINCE PROPOSAL DRAFT



Training window: 2013–2020

Use full SGCharts window. 2020 serotype shift (DENV-3, 35k+ cases) demonstrates concept drift in the report.



Subzone granularity: 330

Fine-grained for operational fogging. Planning areas (55) lose deployment precision.



Weather source: MSS, not NEA API

MSS station data extends usable window back to 2013. NEA API has Dec 2016 floor + 7-hr ingest.



Real-time alert is a rule, not a 2nd model

No outcome-level labels exist publicly. Score $\times$ vulnerability $\times$ count threshold — transparent and honest.